

Quantum challenge: On IIT-DRDO demonstration of quantum key-distribution scheme

Administrative reform is **essential** for India to make advances

Scientists from IIT-Delhi and the DRDO recently **demonstrated** a key advance in quantum cybersecurity that **stands to revolutionise** how the world will communicate in the future. The world's secrets are currently stored and communicated via channels that are protected by difficult mathematical problems. Over the years, an increasing **amount** of computing power available to certain **actors** **has** forced these problems to become difficult and **nigh-uncrackable**. The **imminence** of quantum computing **requires** this **paradigm** to change because quantum computers can (at least on paper) solve problems currently out of reach of the most powerful **conventional** supercomputers. Quantum cybersecurity is one **prong** of this change, promising to protect communication channels **irrespective of** the computing power available to **malicious** actors. The IIT-Delhi and DRDO team successfully demonstrated a quantum key-distribution scheme through one kilometre of free space on the institute's campus. Such a technology allows two individuals (or stations) located a kilometre apart to securely access messages they send each other. If an **eavesdropper** tries to **intercept** any message, **instantaneous** changes in the keys the individuals use to **decrypt** the messages will reveal the channel has been **compromised**, and in a way that the eavesdropper cannot prevent. If **scaled up** to include satellites, the technology could allow stations anywhere in India to exchange information through a quantum network without fear of being compromised.

Herein lies the rub. Quantum communications is one of the four **themes** of the National Quantum Mission, approved in 2023 with an **outlay** of ₹6,003 crore until 2031. **Many** of the same problems **assailing** fundamental research in India **have** already **beset** research under the Mission, however. A small **fraction** of the outlay has been **disbursed thus far** even as **venture capital** flow into startups remains **trivial**. Scientists have complained that **just-in-time** funding, absence of single-window clearances, and documentation requirements have increased the duration of projects. There are **persistent** foreign hardware and software dependencies: **materials** required for specific use-cases, such as **cryostats** and **sensors**, **need** to be **fabricated** abroad while most quantum software **stacks** are currently implemented by multinational companies. Government pay does not match global offers and lack of timely access to resources has forced researchers to accept short-term contracts and rent equipment. In fact, India's **commitment**, itself down from the ₹8,000 crore announced in 2020, **is dwarfed** by those of the U.S. and China, which are five- and 20-times higher, respectively. If the demonstration at IIT-Delhi is to scale, the government cannot simply 'clip on' scientific talent and technological and economic opportunity to **existing infrastructure**. Administrative reform is essential.

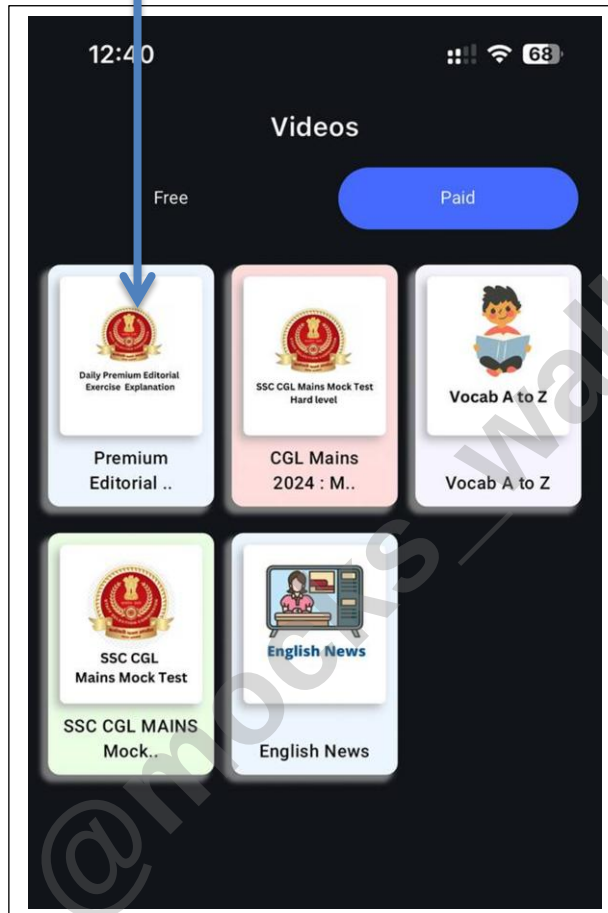
- Red/blue coloring of words in the sentence indicates subject verb relationship; where 'red' denotes 'subject' and 'blue' denotes 'verb'. **[Practice Exercise]**
- **Clip on** (verb) – attach, fasten, fix, connect, latch जोड़ना

Vocabulary

1. **Quantum computing** (noun) – a multidisciplinary field comprising aspects of computer science, physics, and mathematics that utilizes quantum mechanics to solve complex problems faster than on classical computers.
2. **Demonstration** (noun) – display, presentation, exhibition, showcase, proof प्रदर्शन
3. **Essential** (adjective) – necessary, vital, crucial, indispensable, fundamental आवश्यक
4. **Demonstrate** (verb) – show, exhibit, prove, display, illustrate प्रदर्शित करना
5. **Stand** (to) (verb) – be likely to, be expected to, be in position to, seem ready to, appear poised to संभावना होना
6. **Revolutionise** (verb) – transform, overhaul, reshape, innovate, modernize क्रांतिकारी बदलाव लाना
7. **Actor** (noun) – participant, player, entity, stakeholder, agent भागीदार / पात्र
8. **Nigh-uncrackable** (adjective) – almost unbreakable, nearly impossible, tough, resistant, invulnerable लगभग न भेदने योग्य
9. **Imminence** (noun) – closeness, nearness, approach, anticipation, impending arrival आसन्नता
10. **Paradigm** (noun) – model, framework, pattern, structure, approach ढांचा
11. **Conventional** (adjective) – traditional, standard, usual, established, normal पारंपरिक
12. **Prong** (noun) – aspect, element, part, branch, phase पहलू
13. **Irrespective of** (phrase) – regardless of, despite, even though, no matter, बिना ध्यान दिए
14. **Malicious** (adjective) – harmful, hostile, spiteful, wicked, malevolent दुर्भावनापूर्ण
15. **Eavesdropper** (noun) – spy, snooper, listener, interceptor, intruder बात चुराकर सुनने वाला
16. **Intercept** (verb) – catch, block, interrupt, seize, prevent रोकना
17. **Instantaneous** (adjective) – immediate, quick, prompt, sudden, rapid तुरंत
18. **Decrypt** (verb) – decode, unscramble, unlock, interpret, translate कोड खोलना
19. **Compromise** (verb) – breach, weaken, undermine, jeopardize, violate जोखिम में डालना
20. **Scale up** (phrasal verb) – expand, grow, increase, enlarge, amplify बढ़ाना

21. **Herein lies the rub** (phrase) – this is the problem, here is the issue, the difficulty is, this is the catch यही असली समस्या है
22. **Theme** (noun) – subject, topic, idea, concept, focus विषय
23. **Outlay** (noun) – expenditure, budget, allocation, funding, spending व्यय / बजट
24. **Assail** (verb) – trouble, plague, burden, attack, afflict हमला करना / परेशान करना
25. **Beset** (verb) – overwhelm, encircle, burden, plague, surround घेरे रहना
26. **Fraction** (noun) – portion, segment, part, sliver, piece अंश
27. **Disburse** (verb) – pay, allocate, distribute, spend, grant भुगतान करना
28. **Thus far** (phrase) – until now, so far, to date, up to this point अब तक
29. **Venture capital** (noun) – capital invested in a project in which there is a substantial element of risk, typically a new or expanding business. उद्यम पूंजी
30. **Trivial** (adjective) – unimportant, insignificant, minor, negligible, petty तुच्छ
31. **Just-in-time** (adjective) – immediate, last-minute, on-demand, real-time, near-deadline समय पर
32. **Persistent** (adjective) – continuous, constant, unrelenting, ongoing, enduring लगातार
33. **Cryostat** (noun) – an apparatus for maintaining a very low temperature. ठंडा रखने वाला यंत्र
34. **Sensor** (noun) – detector, device, gauge, indicator, monitor संवेदक
35. **Fabricated** (adjective) – manufactured, constructed, produced, built, assembled निर्मित
36. **Stack** (noun) – the levels of programming, software, etc. used in a mobile app or website, in order to create the app or website or keep it working कंप्यूटर सॉफ्टवेयर की परतें
37. **Commitment** (noun) – dedication, promise, pledge, obligation, resolve प्रतिबद्धता
38. **Dwarf** (verb) – overshadow, outshine, surpass, reduce, minimize छोटा दिखाना
39. **Existing** (adjective) – current, present, ongoing, established, prevailing वर्तमान
40. **Infrastructure** (noun) – foundation, framework, system, base, structure आधारभूत संरचना

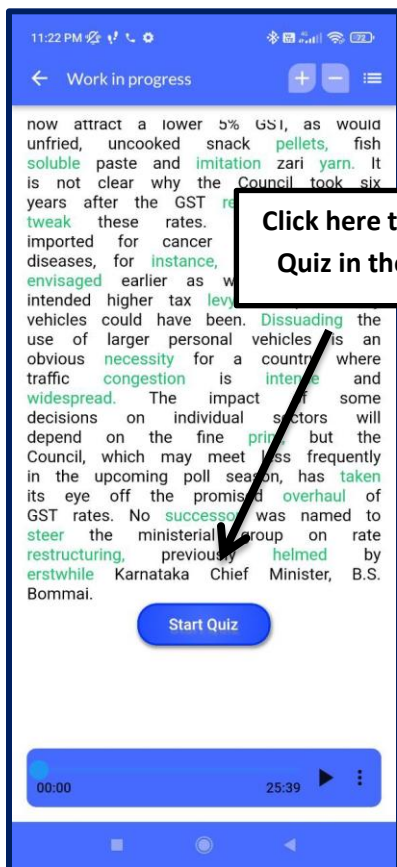
Video solution of Practice Exercise from this Editorial is exclusively available on App.



Summary of the Editorial

1. **Key Achievement:** Scientists from IIT-Delhi and DRDO demonstrated a significant quantum cybersecurity advance using quantum key-distribution (QKD) technology.
2. **QKD Demo:** They securely transmitted data over one kilometre of free space, showcasing a future-ready communication method.
3. **Quantum Security Advantage:** QKD ensures that any interception attempt by a third party changes the key, alerting users to a breach instantly.
4. **Future Potential:** When scaled with satellite integration, QKD could enable secure communication across India via a quantum network.
5. **Urgency Due to Quantum Computing:** Quantum computers could break existing encryption methods, necessitating quantum-safe alternatives.
6. **National Quantum Mission:** Approved in 2023 with a ₹6,003 crore budget, it aims to advance quantum technologies, including communications.
7. **Administrative Bottlenecks:** Despite the mission's promise, only a small portion of the allocated funds has been disbursed.
8. **Startup Challenges:** Venture capital investment in Indian quantum tech startups is negligible, limiting innovation and growth.
9. **Funding and Bureaucracy Issues:** Scientists face delays due to just-in-time funding, excessive documentation, and lack of single-window project clearances.
10. **Foreign Dependence:** India relies on foreign-made hardware like cryostats and sensors and foreign-developed quantum software.
11. **Human Resource Challenges:** Low government salaries and contract-based research jobs drive talent to short-term arrangements or abroad.
12. **Infrastructure Inadequacy:** Researchers often rent expensive equipment due to a lack of institutional support and timely access to tools.
13. **Reduced Commitment:** The original ₹8,000 crore commitment has been reduced, and still lags behind U.S. and China's massive investments.
14. **Global Comparison:** India's spending is five times lower than the U.S. and 20 times less than China's in the quantum technology domain.
15. **Need for Reform:** For quantum technologies to scale and thrive, India must implement deep **administrative reforms** and not merely rely on existing infrastructure.

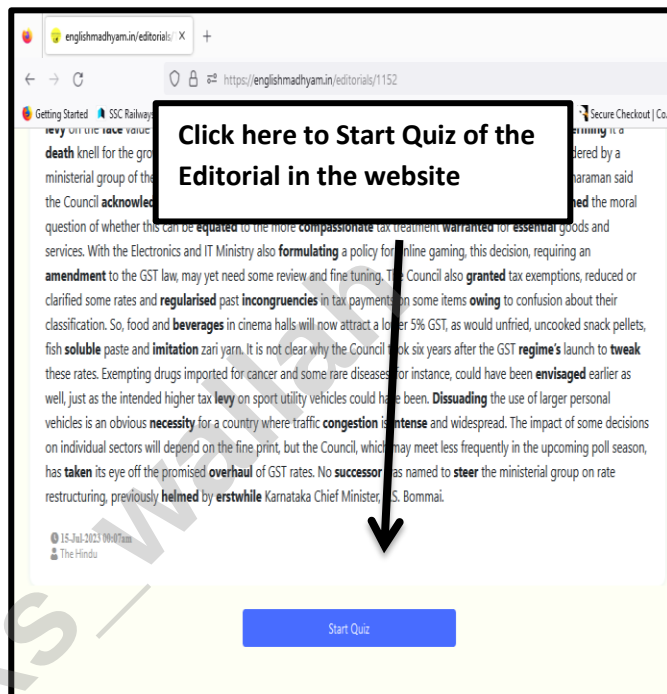
Practice Exercise: SSC Pattern Based



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